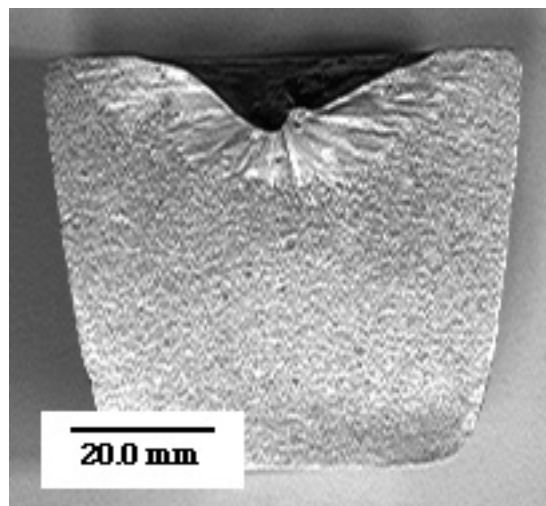


NPRE 432: Engineering Materials Laboratory Lab 4 (from ME 330)

Phase Diagrams and Casting



0.2 wt% Cu in Al, 5 °C superheat, graphite mold

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I. Introduction

This experiment explores the macrostructure and demonstrates the temperature history encountered during solidification and its utility in the construction of phase diagrams. Phase diagrams are constructed experimentally from solidification cooling curves, which are simply temperature versus time plots. A bismuth–tin phase diagram will be constructed from several cooling curves of various compositions. In addition, several different aluminum castings will be examined.

The objectives of this laboratory experiment are:

- (1) Construct a phase diagram for bismuth–tin solution using experimentally obtained cooling data;
- (2) Understand the use of phase diagrams;
- (3) Explore the macrostructure and microstructure of cast materials;
- (4) Explore the impact of solidification on casting quality.

A. Phase and Components

A phase is defined as a homogeneous portion of matter. This definition implies that different regions of a phase have essentially the same chemical composition, atomic arrangement, and physical properties. For example, the three phases of water are vapor, liquid, and ice. Figure 1 shows that a mixture of liquid water and ice is actually two phases.

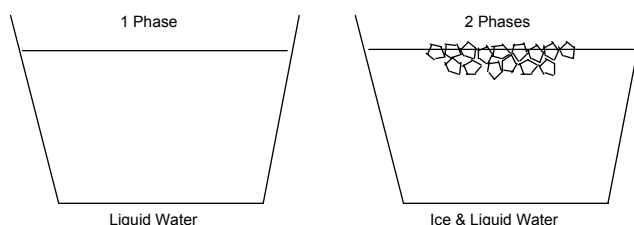


Figure 1: Definition of a Phase

Mixtures of salt (NaCl) and water provide another common example. If a small amount of salt is added to water at room temperature, all of the salt will dissolve to form a solution called brine (Fig. 2a). If salt is continuously added to this mixture, eventually the solution becomes saturated and any further salt added will remain as solid NaCl (Fig. 2b). Alternatively, if we start with brine at room temperature, and then lower the temperature sufficiently, some of the water will freeze, forming a mixture of ice and brine (Fig. 2c).

Although a polycrystalline pure metal has many grains, it is still considered a single phase. The grains may differ in size, shape, and orientation, but the composition of each grain is

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identical. If sufficient alloy is added to the metal, then it may precipitate a separate phase(s) in a similar manner to the salt or ice crystals in brine.

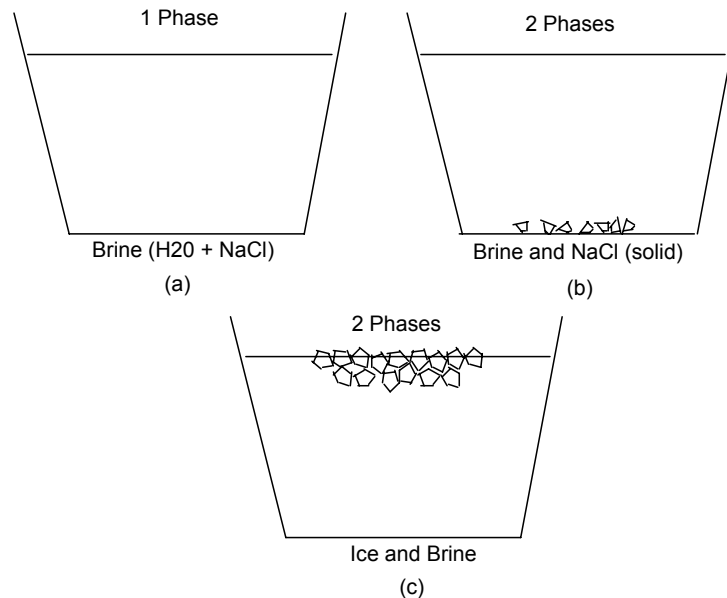


Figure 2: Phase and Component Differences

II. Cooling Curves and Phase Diagrams

Phase diagrams are often called "equilibrium diagrams" since they show the phases present in a system of a given composition and temperature at equilibrium. In other words, the cooling process is slow enough that all possible phase changes have occurred at any given temperature. Phase diagrams are constructed from cooling curves for a given alloy composition by noting the temperatures at which the slope of the curve changes. Another approach to constructing a phase diagram requires observing the microstructure of the mixture under a "hot stage" microscope as the metal cools. This technique is considerably more time consuming and cost intensive since such microscopes can be quite expensive.

A. Pure Metals

When a pure liquid metal cools, it reaches the melting point (a unique temperature) and solidifies. A cooling curve (Fig. 3) for this situation reveals that at the point of solidification, the temperature remains constant (zero slope). The energy that is liberated during solidification is not reflected as a temperature change; rather it is supplied by the "latent heat" of solidification. Additionally, the liquid and solid have different thermal conductivities due to differences in their atomic organization. Thus, the slopes of the cooling curve for each region (liquid and solid) is

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usually different. Note that the temperature drops slightly below the equilibrium freezing temperature before solidification begins. This phenomenon is called “supercooling” and is due to the need to establish a thermodynamic driving force for solidification.

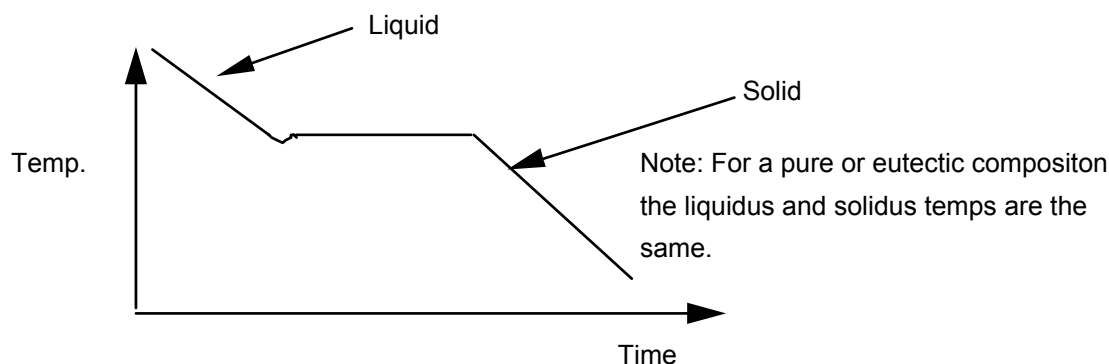


Figure 3: Solidification Cooling Curve for a Pure Metal

B. Metal Alloys

In cooling a metal mixture, the process is more complex. Typically the metal will cool through a two-phase region, where solid and liquid coexist over a range of temperatures. This is called the "mushy" stage, and is similar to salt-water "slush." Instead of a constant temperature during solidification as in a pure metal, the cooling curve continues to have a negative slope. This region is usually non-linear due to evolution of latent heat as the percentage of solid and liquid changes during solidification. The temperature at which the slope first changes corresponds to the first appearance of solid and is called the liquidus temperature. After solidification is completed, the slope changes again. The temperature at which this occurs corresponds to the point of complete solidification and is called the solidus temperature. A phase diagram can be constructed by recording these cooling curves for various compositions and plotting the liquidus and solidus temperatures versus alloy composition.

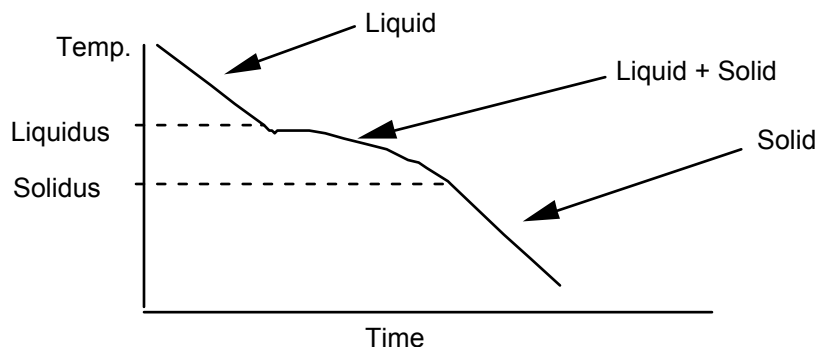


Figure 4: Solidification Cooling Curve for a Metal Alloy

It should be noted that the liquidus and solidus temperatures are equal for a pure metal and an alloy of eutectic composition. A eutectic reaction is one in which a liquid transforms directly into two different solids during cooling without going through a "mushy" stage. Further phase transformations and multi-phase regions may be encountered at lower temperatures.

III. Quantitative Phase Analysis

Using a phase diagram, it is possible to evaluate important aspects of the physical structure of the metal and the chemical composition of the constituent phases.

A. Phase Compositions in a Two-Phase Region

For a given alloy cooling through a two-phase field, a tie line can be constructed by connecting the two boundaries of the field at constant temperature. This line has been drawn in Fig. 5 for a certain temperature. The two phases present (e.g. α and L) are indicated by the phases present in the adjacent fields. The compositions of the phases present are the compositions at the point the tie line meets the boundaries of the field. These points are labeled b and c in Fig. 5. In general, the composition of the individual phases changes as the mixture is cooled through a two-phase field even though the average composition of the mixture is constant (point a). Note that the caption in Fig. 5 refers to a "binary" phase diagram. This terminology means that there are just two components for the given material (element A and element B in this case). Steel, for example, has many different alloying elements, and thus is not binary.

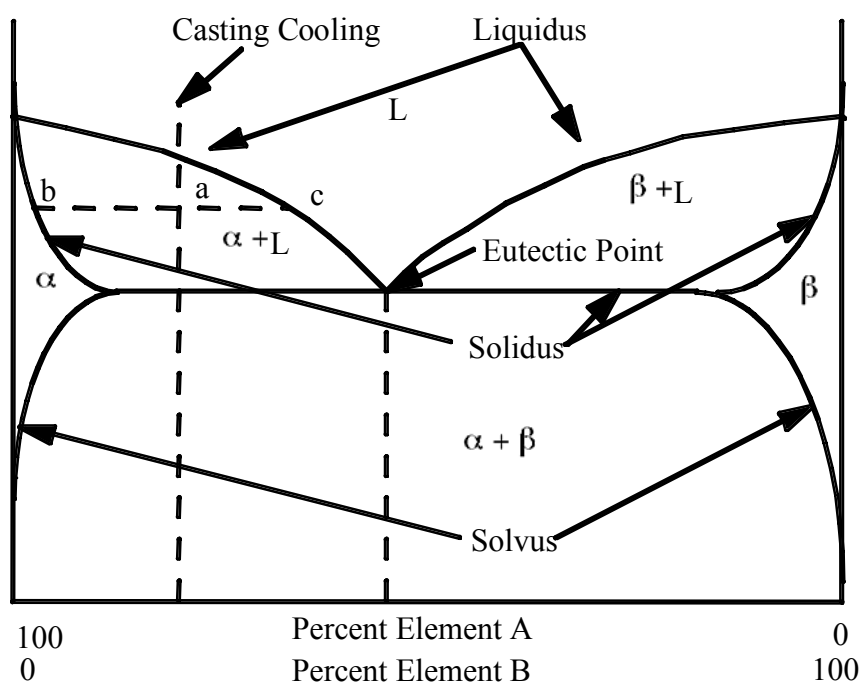


Figure 5: Generic Binary Eutectic Phase Diagram

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B. Amounts of Phases

The method used for determining amounts of phases in equilibrium is commonly called the Inverse Lever Rule. Drawing a tie line, three compositions can be identified: the composition of the mixture, and the compositions given by the tie line at the intersection with the two boundaries of the field. These points are labeled a, b, and c respectively in Fig. 5 for a particular alloy composition. The Inverse Lever Rule states that the amount of the phase on the left side (α phase) is proportional to the difference in compositions on the right side of the mixture (from point a to point c) relative to the total distance from b to c. In Fig. 5, the percent of the α phase is given by:

$$\% \alpha = \frac{ac}{cb} \times 100 \quad (1)$$

where ac and cb refer to differences in composition along the tie line. The quantity cb is simply the composition at c minus the composition at b. The percent liquid %L is:

$$\% L = \frac{ab}{cb} \times 100 \quad (2)$$

IV. Casting

A wide variety of products are made by casting, which is accomplished by pouring molten metal into a mold. Although everyone is familiar with cast products, it is not widely appreciated that most man-made objects involve solidification at some stage. Phases present in the cast products can largely be predicted from the phase diagram. However, morphology of the microstructure will depend on the process variables such as cooling rate, superheat, vibration, etc.

Molds for castings can be either permanent (e.g. steel dies) or temporary (e.g. sand mold). In sand casting, the mold is reconstructed for each new casting. This facilitates easy removal of the casting from the mold, and the sand is reusable. The mold is formed around a pattern, which replicates the shape the final piece is to have. However, the pattern is scaled larger than the intended product to account for shrinkage in the mold. As would be expected, various metals shrink different amounts. Thus, pattern making is itself a skilled art.

A schematic illustration of a sand mold is shown in Fig. 6. The sand in the mold is retained in a steel ring called a flask. The lower mold is called the drag and the upper mold is the cope. The parting line is the seam separating the two halves of the mold. The molding sand is graded based on the size range of the sand particles used in a mold. Only one grade of sand is used for a mold. The sand is held together with a coating of bentonite clay, wetted with water or oil. This mixture is called "green sand," where green refers to the fact that the sand contains water. Green strength is a commonly used term that refers to the sand's ability to hold together in the mold.

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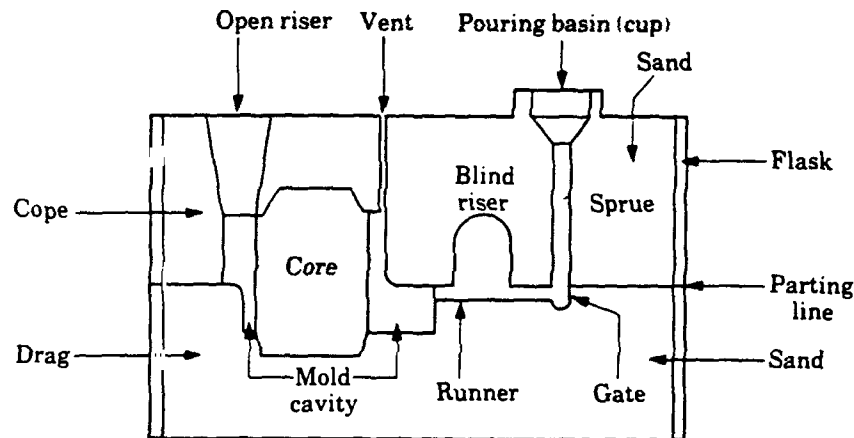


Figure 6: Schematic Illustration of a Sand Mold

The lower half of the pattern is placed in the drag flask, which is loosely filled with sand. The pattern is dusted with a parting agent (usually silica or bone) to allow easy removal from the sand. The sand is then rammed, which refers to packing of the sand to hold its shape after the pattern is removed. Ramming by machine is often called jolting. In some production processes the sand is dumped into the flask such that it is rammed by its own weight. The cope flask is then placed over the drag, filled with sand, and rammed. The two halves are split again and the pattern is removed. The line between the cope and drag defines the parting line. Match-plate patterns have cope and drag portions on opposite sides of a single plate, so the two mold halves can be rammed simultaneously.

A channel, called a gate forms a horizontal conduit for introducing molten metal into the mold from the vertical sprue. Risers are used to provide feed metal to compensate for shrinkage. For some simple castings, a single channel can serve as both gate and riser. For large castings, or those with complex geometries, several gates and risers may be necessary to ensure quality. The location of the gate has a large influence on mold filling and solidification. Gating is planned in advance, and many patterns include both gates and risers.

The function of the riser is to compensate for shrinkage, since there is a decrease in volume as the metal solidifies. An important consideration in casting design is assuring that shrinkage does not result in cavities, tearing, or incomplete mold filling. A common approach is to maintain a head of liquid in the riser, allowing the liquid to continue flowing into the mold as the casting shrinks. The surface of the riser, which is later removed from the finished casting, will have a hollow center due to shrinkage.

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When an internal cavity is required, a mold insert called a core is placed in the mold during pouring. The core is formed separately from the cope and drag, then positioned within the mold prior to pouring. A core box is used as the pattern to form a core.

Another method for creating the mold is investment casting. A wax or polymer model of the component is constructed, and a liquid cement-like slurry (the investment) is poured around the mold. The mold is heated, which melts the model, and liquid metal is added to the mold. Unlike sand, investment material is typically not reusable.

During a casting process, the metal is typically poured at a temperature above its melting point to assure proper mold filling. The energy represented by this raised temperature is called superheat. The superheat must be removed by cooling the liquid before the casting can solidify. After the metal has solidified, the sand is removed from the mold and literally shaken apart to retrieve the casting. The sand is recovered for reuse, but quality checks of properties such as green strength are necessary to maintain consistent performance.

V. Solidification Macrostructures

The as-cast structure is very influential on the final properties of the casting. Low-magnification examination of etched casting sections and fracture surfaces can provide valuable information about cast materials. The grain size and shape (ranging from equiaxed to long, thin columnar grains) are both affected by the alloy composition and casting conditions. However, the grain structure must first be examined before the effects of casting variables can be considered. This is done through careful polishing and etching of sample specimens, which is described in the procedure section.

A. Microstructures and Casting Zones

In general, castings have three distinct zones as seen in Fig. 7:

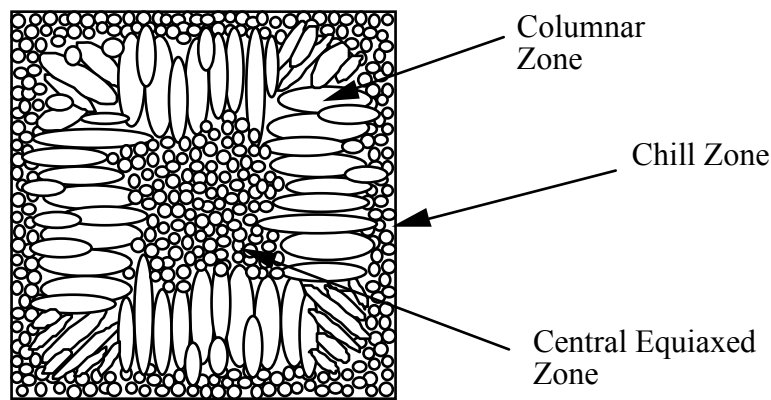


Figure 7: Typical Zones Found In a Cast Material

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- (1) The chill zone is found at the surface of the mold. It arises from rapid cooling of the liquid metal which nucleates a thin layer of fine, equiaxed grains. The mold surface also provides many nucleation sites for grain formation.
- (2) The columnar zone is composed of long, thin grains aligned in the direction of heat flow. For this simple mold shape, the direction is perpendicular to the mold wall. Only grains that are favorably oriented will grow toward the center of the solidifying mass. These are formed by dendrites that form and grow under ordinary cooling conditions. In some of the columnar-grained specimens in the laboratory, it is possible to see the leaved dendrites. This columnar structure is undesirable because the parallel grain boundaries weaken the casting.
- (3) A central equiaxed zone, where nucleation and growth of new crystals limits continued growth of the columnar zone (dendrites).

It is important to note that the size and locations of these zones can vary between castings. For example, for a pure metal, the columnar zone extends to the center of the casting and does not form a central equiaxed zone. For an alloy, the formation of this zone is limited by the central equiaxed zone. The equiaxed zone can be extended throughout the casting (excluding the chill zone) by the addition of grain-refining agents, which induces nucleation of grains. This effect is enhanced by reducing or eliminating thermal gradients within the melt. An equiaxed grain is simply a grain that is roughly the same size in all directions. Dendrites form a leafy or feathery structure because crystals have preferred growth directions. For example, cubic crystals grow fastest in {100} directions.

B. Grain Size

The grain size in a casting is determined in the same manner as in finer grained specimens. There are several approaches to determining grain size:

- (1) Choose a field of grains using a low-power microscope, count the number of grains in the field, and compute the number of grains per unit area.
- (2) Estimate the average grain diameter by counting the number of grain boundaries intersected by a line of known length.
- (3) Make a series of direct measurements of single grains.

The ASTM grain size number is an arbitrary scale given by:

$$\text{ASTM No.} = -6.644 \log (d) - 2.95 \quad (3)$$

where d is the grain diameter in millimeters.

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This number is always stated as an integer, and is a measure of the number of grains per unit volume in the specimen. Of course, if the grains are not equiaxed, the grain size is not a single value. Long, narrow grains obviously have at least two important dimensions. For these structures, the spacing between adjacent dendrite arms is used as a length scale.

VI. Controlling the Structure

There are several aspects of the casting design and foundry technique which can influence the structure produced in the casting. The resulting microstructure has a profound influence on the final properties of the cast component. Conditions that affect the final microstructure include cooling rate, composition, vibration, and superheat. These effects will be examined by comparing several aluminum castings previously prepared in a foundry.

A. Cooling Rate

In general, faster cooling produces a finer grain size by providing a larger thermodynamic driving force for nucleation. Such a cooling rate promotes rapid nucleation of grains, and the large number of grains limits their growth. In a casting, an average grain size of less than 1 mm is considered a fine grained structure. Fine grains are desirable for several reasons. Recall that finer-grained materials are stronger. Finer structures are typically more ductile, and defects such as inclusions, microsegregation, and porosity tend to be more evenly distributed.

The most common method of controlling the cooling rate is to use a different molding material. Sand castings cool slowly compared to other mold materials since cooling occurs primarily through conduction in the sand (which has a low thermal conductivity). Permanent molds such as metal and graphite have higher cooling rates. The cooling rate may be further increased by water-cooling the exterior of the mold. However, steps to increase cooling generally increase the cost of the casting.

B. Composition

Pure metals tend to have few nucleation sites because the composition of the melt is uniform, which produces large columnar zones in the casting. The addition of alloying elements widens the freezing range of the casting and enhances the formation of equiaxed grains. Other additives are grain-refining agents because they form clusters which act as sites (heterogeneous nucleation). Examples of these agents are small wt.% of Ti or Cu in an aluminum castings. Copper as an alloying element widens the freezing range of aluminum, while titanium is strictly a grain-refining agent.

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C. Vibration

Stirring or vibration of the liquid as the casting cools creates currents within the liquid. As dendrites form and grow (which would ordinarily become the columnar zone), they are detached by the currents and fall to the bottom of the liquid. This has two effects: the dendrite fragments become sites for equiaxed grain growth; and the sites where they detached nucleate more dendrites, allowing the process to repeat.

Stirring does not need to be mechanical. Just as metals can be melted under a steady magnetic field (an induction furnace), metals can be stirred using a fluctuating magnetic field. Electromagnetic stirring has been used to improve quality in continuous casting.

D. Superheat

The amount the melt is heated above the liquidus is called superheat. Increasing superheat delays formation of equiaxed grains because uniform grains cannot form until the liquid reaches the solidus temperature. Thus, superheat promotes formation of a larger columnar zone. However, superheat is necessary to assure complete mold filling. The greater the complexity of the casting, the greater the required superheat. Higher temperatures also favor grain growth mechanisms and suppress nucleation.

VII. Experiments

SAFETY WARNING: *A high temperature furnace and molten metal will be used in this experiment. Serious burns will result if you touch the furnace or crucibles without protective equipment. Use tongs and heat resistant insulated gloves when handling the crucibles. Be sure to wear safety glasses, and follow the procedure demonstrated by your TA or lab staff.*

A. Phase Diagram Equipment

Ideally a calorimeter with a very slow constant rate of heat extraction would be utilized to obtain the cooling curves needed to construct a phase diagram. This is impractical for a 2 hour lab session. Instead, air-cooling of an alumina crucible (3.0 W/m-K) will be utilized. While this simpler cooling methodology has drawbacks for determining solid-solid phase transitions, it works rather well for the higher temperature liquid-solid transformations. A thermocouple enclosed in a stainless steel probe is used to determine the temperature of the bismuth-tin melt. A PC-based digital data acquisition system displays and stores the time-temperature data. The overall procedure illustrates the important concepts relevant to the construction of a bismuth-tin phase diagram.

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B. Phase Diagram Procedure

Research quality phase diagrams are generated with very high purity (99.999% pure) constituents, which are quite costly even for basic materials such as tin and bismuth. This experiment utilizes 99% to 99.9% pure materials. This will alter the temperatures observed, especially for the 100% compositions, but the characteristics of the curves will allow construction of portions of the phase diagram. Each lab section will generate data for 8 of the 16 compositions listed in Table 1.

- (1) Bismuth and tin (formed as shot or needles) are available in the lab. Each half of the lab section will mix one of the wt.% listed in Table 1 as well as the lab data sheet. (Total weight of the sample is 100g.) After weighing each material (for the composition assigned to your group) in the plastic cups provided, put the material into a crucible.
- (2) Place the crucible into the furnace, which is maintained at 350°C. This high temperature can cause severe burns, so please use caution. Use the long tongs to handle the crucible, and heat resistant gloves to open the furnace. *As always, everyone must be wearing safety glasses.*
- (3) There will be other Bi-Sn combinations in the furnace. Each group will be assigned two compositions for testing, one of which may be the one that you mixed. Data for other compositions will be available.
- (4) Set up the data acquisition system per instructions from your TA.
- (5) With an empty crucible practice the procedure of i] opening the furnace, ii] placing the crucible on the target area of the heat resistant brick, and iii] lowering the thermocouple approximately 2 to 5 mm above the bottom of the crucible (first lower it to the bottom, and then raise it slightly so it is not touching the crucible bottom surface).
- (6) After you are comfortable with the procedure and the group members are all synchronized, remove a crucible from the furnace and generate a portion of the cooling curve by acquiring temperature data as the mixture cools. Be sure not to start with the sample you just put in the furnace!
- (7) After completion of both cooling curves, enter your group's data into the class computer data file. Data from other sections will allow completion of Table 1.

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Table 1: Melt Composition and Cooling Curve Data Sheet

Wt. % Bi	Inflection Temperatures (°C) / time (sec)			Comments
0	/	/	/	
5	/	/	/	
10	/	/	/	
15	/	/	/	
20	/	/	/	
25	/	/	/	
30	/	/	/	
40	/	/	/	
50	/	/	/	
55	/	/	/	
60	/	/	/	
65	/	/	/	
70	/	/	/	
80	/	/	/	
90	/	/	/	
100	/	/	/	

C. Castings

- (1) A short video clip (~12 min.) of different casting operations will be shown in the classroom. Examples of casting molds are available for viewing. Please sketch half the mold master and label the pertinent features.
- (2) In the lab, two specimens of the type shown in Figure 8 are available for observation. One of the specimens was gated at the thin end, the other at the thick end. Observe and sketch the gating samples, noting the shrinkage cavity locations and pour direction on Figure 8.
- (3) Also observe other castings with and without defects.

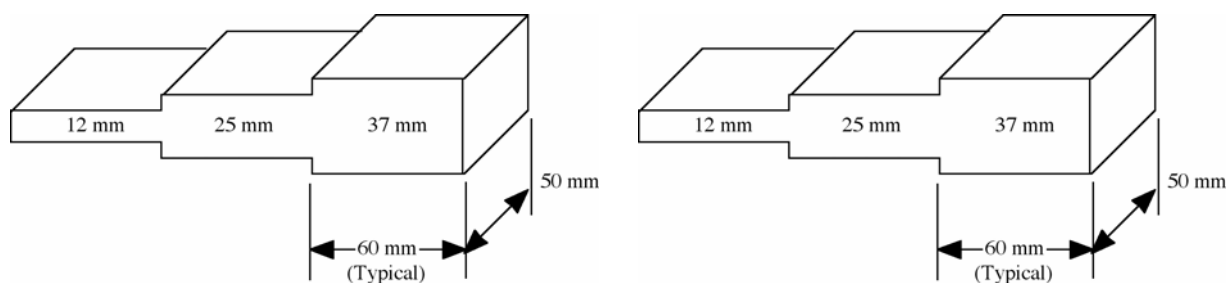


Figure 8: Casting Specimen

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(4) Four aluminum alloys were used for the casting samples to be studied in this experiment. The details of each of the numbered specimens are shown in Table 2. Samples of the aluminum castings described in Table 2 have been cut in half and metallographically polished. A variant of Tucker's etch was used (15 ml HCl, 5 ml HNO₃, 5 ml HF, 75 ml H₂O) to reveal the macrostructure. Observe the macrostructures and label the photos shown in the Appendix with composition, other information and sample # from Table 2. Table 3 is provided to assist the student in following the sequence in which they are displayed on the lab boards (put sample # corresponding to compositions, mold type and superheat in the grids provided. Some sample #s may repeat!). Use of the illuminated magnifiers may be helpful, especially in the equiaxed regions. A photograph of the equiaxed grain structure of a 0.04wt.% Ti-aluminum casting (specimen #8) is shown in Figure 9. *Please avoid putting fingerprints or any other marks on any polished sample—it takes considerable time to polish them!*

Table 2: Casting Samples

Sample #	Composition	Mold Type	Superheat	Comments
1	99.9% Al	sand	70°C	vibrated mold
2	1.5% Cu in Al	sand	50°C	
3	0.04% Ti in Al	sand	40°C	
4	99.9% Al	graphite	50°C	
5	0.2% Cu in Al	graphite	50°C	
6	0.2% Cu in Al	graphite	50°C	
7	1.5% Cu in Al	graphite	50°C	water cooled mold
8	0.04% Ti in Al	graphite	40°C	
9	99.9% Al	graphite	110°C	
11	99.9% Al	graphite	110°C	cover photo
12	1.5% Cu in Al	graphite	70°C	
13	0.2% Cu in Al	graphite	5°C	

Table 3: Casting Sequence Charts

Effect of Mold Cooling		Effect of Changing Alloy	

	Effect of Superheat		
	Effect of Composition		
Graphite Mold			
Sand mold			

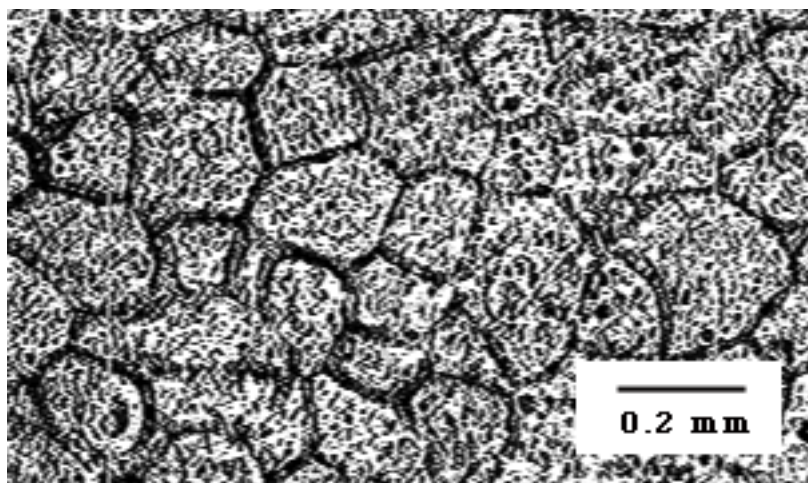


Figure 9: 0.04wt.% Ti-aluminum casting - Equiaxed Grain Structure

D. Cast Iron Metallography

View the gray iron and nodular cast iron (etched with a 1% Nital solution: 1 ml HNO₃ in 99 ml methanol) in the optical microscope. Measure either 5 graphite flakes or nodules at the appropriate magnification with the metallographic measurement system. Make sure that the scale factor and magnification correspond. These data will be used to categorize the graphite in the cast iron according to the classification data in Table 3.

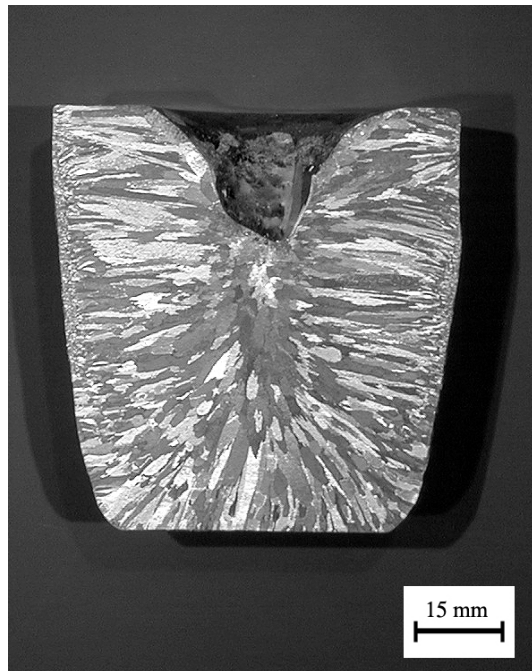
Table 4: Graphite size designation

Size Class	X _{max} (mm)
1	0.128
2	0.064
3	0.032
4	0.016
5	0.008
6	0.004
7	0.002
8	0.001

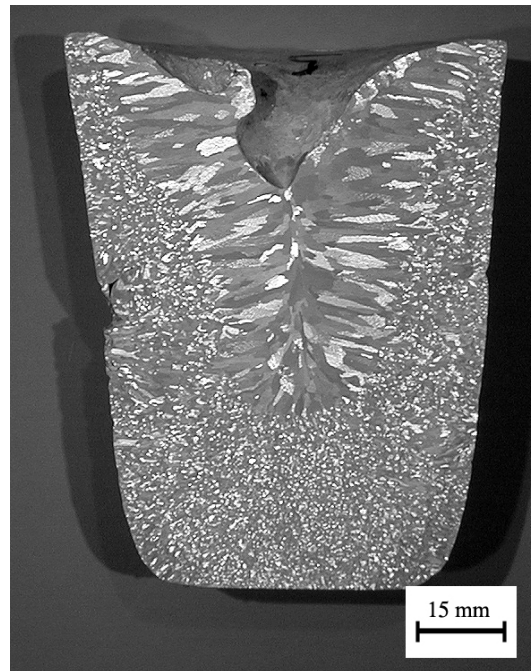
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Appendix—Aluminum Casting Macrostructures



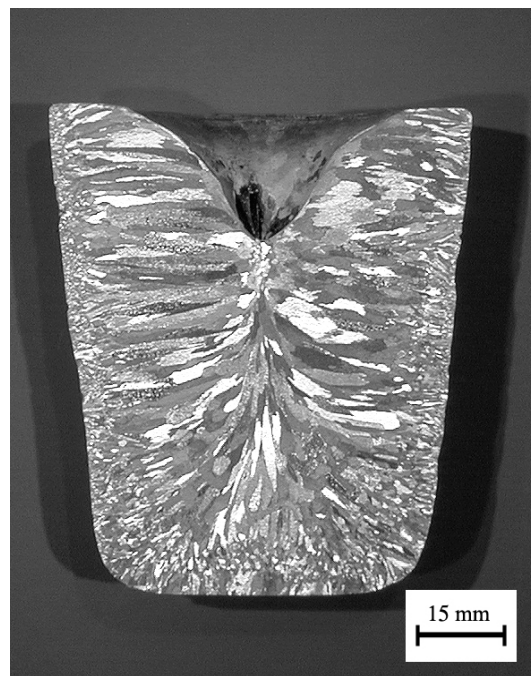
Composition: _____
Mold type: _____
Superheat: _____



Composition: _____
Mold Type: _____
Superheat: _____

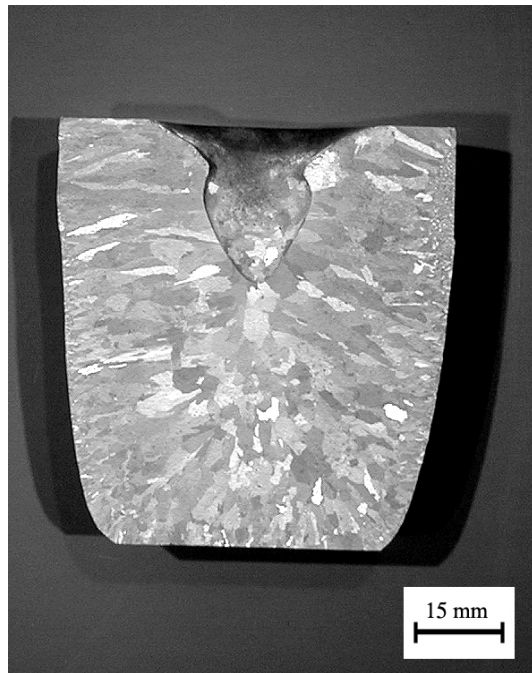


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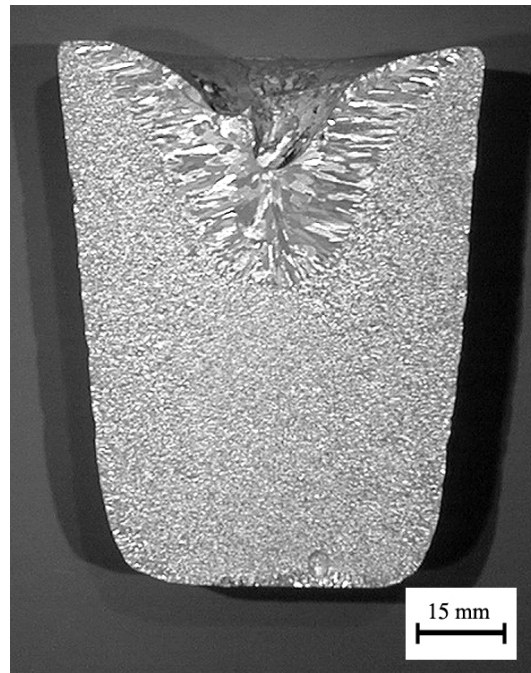


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Mold Type: _____
Superheat: _____

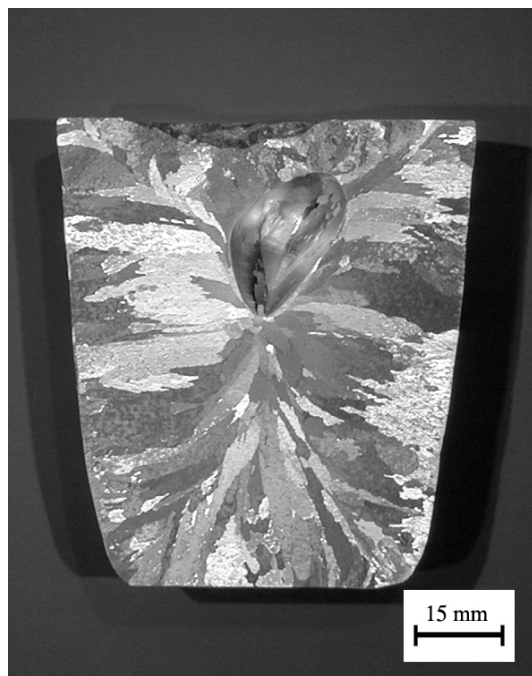
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Appendix—Aluminum Casting Macrostructures



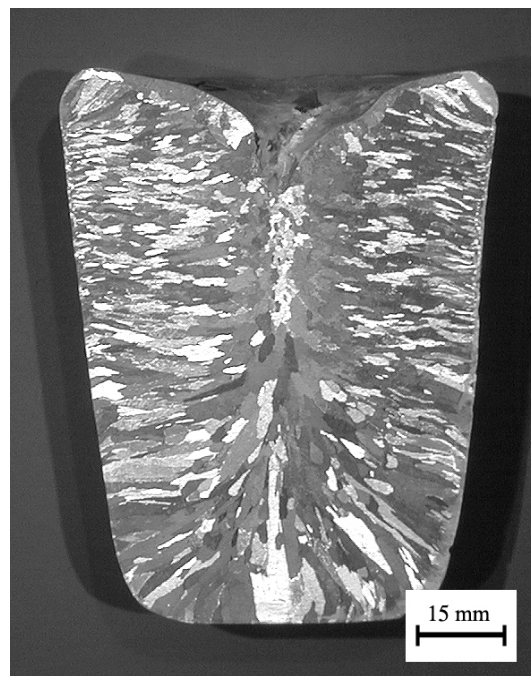
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Mold type: _____
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Composition: _____
Mold Type: _____
Superheat: _____

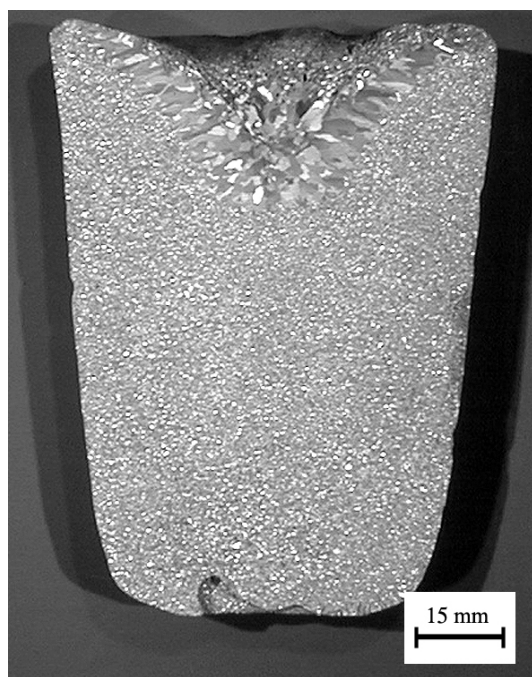


Composition: _____
Mold type: _____
Superheat: _____

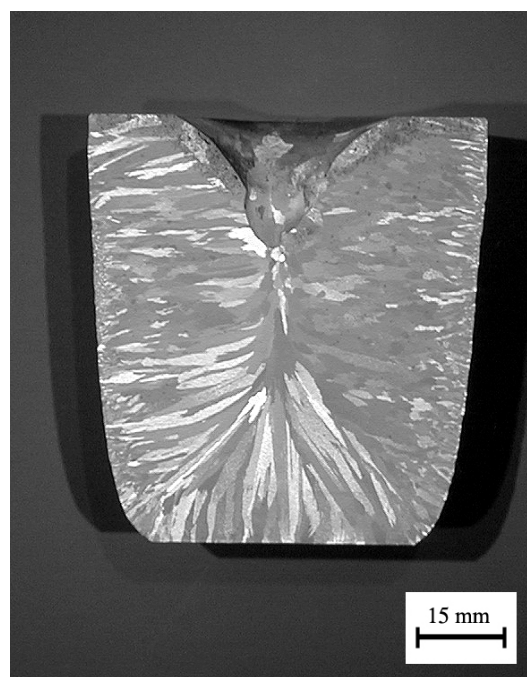


Composition: _____
Mold Type: _____
Superheat: _____

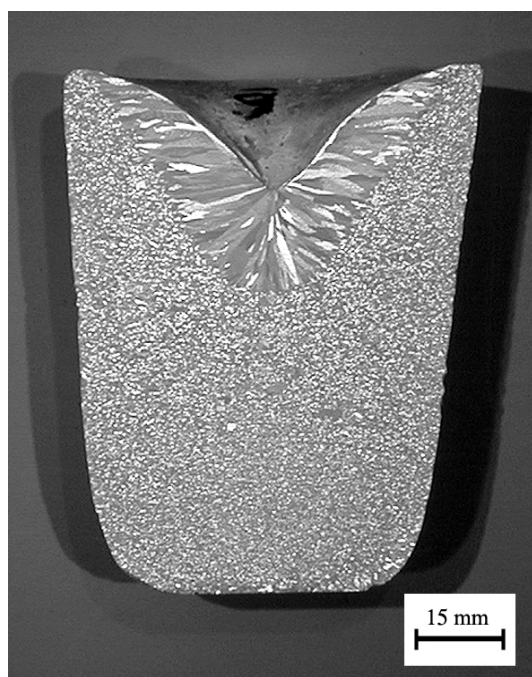
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Composition: _____
Mold type: _____
Superheat: _____



Composition: _____
Mold Type: _____
Superheat: _____



Composition: _____
Mold type: _____
Superheat: _____



Composition: _____
Mold Type: _____
Superheat: _____

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Appendix—Aluminum Casting Macrostructures

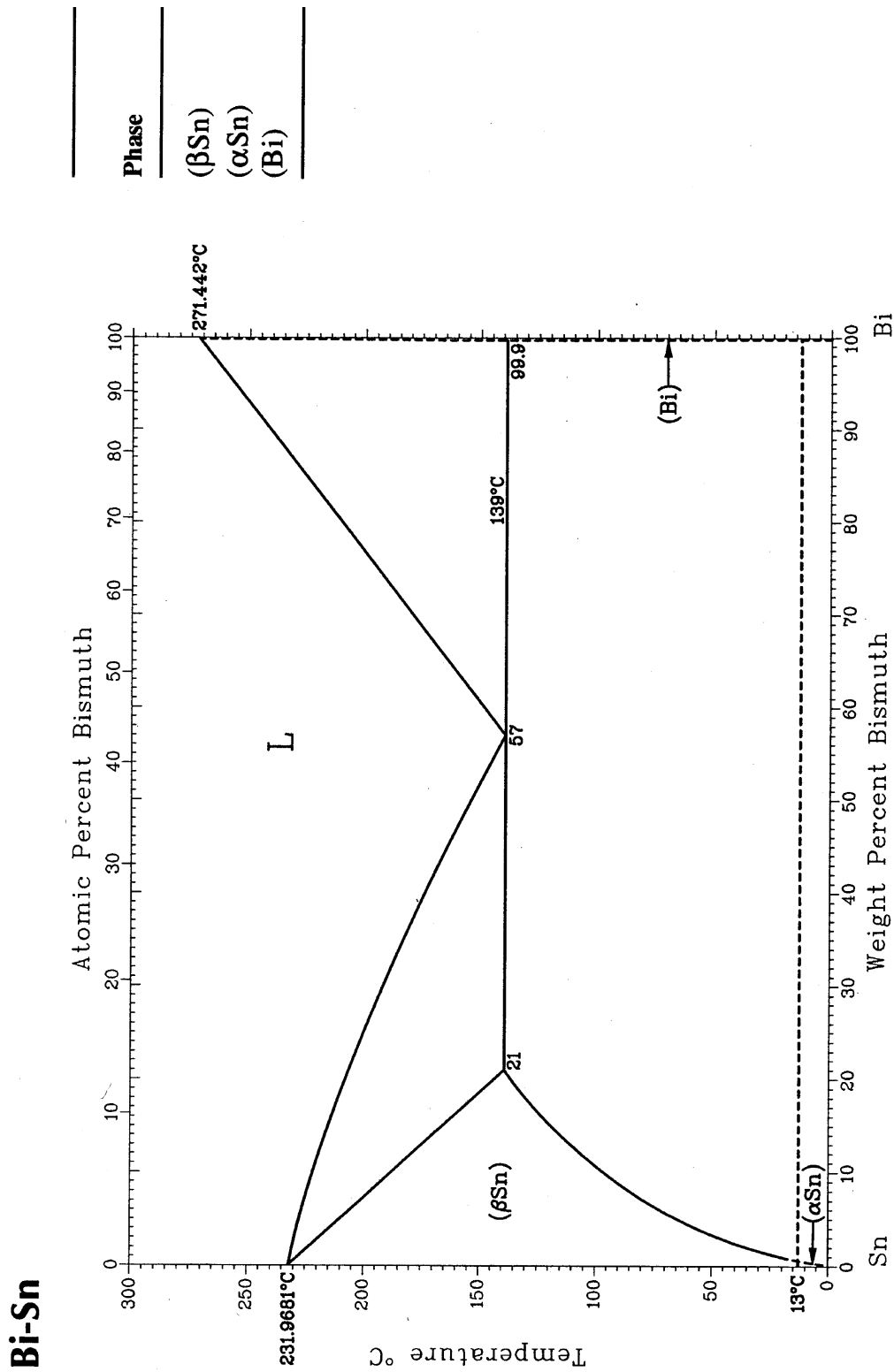


Figure 10: Bismuth-Tin Phase Diagram (Ref: *ASM Handbook*, Vol. 3)